CUSTOMER SEGMENTATION

Internship Project Report

A data analytics and unsupervised machine learning project using Python, Pandas and Scikit-learn.

Dataset disclosure: This report uses 1,200 synthetically generated customer records for demonstration and learning. It is not actual Thiranex or employer customer data. Replace it with authorised, anonymised business data before drawing real business conclusions.

1. Executive summary

This project groups customers with similar demographic and purchasing characteristics using K-Means clustering. The workflow includes data preparation, feature scaling, cluster-count comparison, customer-level segment assignment, visualisation and business recommendations.

2. Business objective

The objectives are to identify groups with similar purchase behaviour, compare spending and purchase patterns across groups, and recommend targeted retention, loyalty and personalisation actions.

3. Dataset and features

The sample contains 1,200 customer records. Clustering variables: age, annual income, spending score, purchases in the last 12 months, average order value, days since last purchase, monthly website visits and discount usage rate. Gender, city tier and membership are included as descriptive attributes but are not used directly in clustering.

4. Methodology

The CSV is loaded with Pandas. Missing numeric values are handled using median imputation. StandardScaler normalises the numerical features. K-Means is tested for k=2 through k=6, and silhouette scores are compared. The best-scoring k is selected. PCA is used to project the scaled data into two dimensions for visual inspection.

5. Model selection

The selected model uses the number of clusters with the highest silhouette score among the tested values. Silhouette score measures how well-separated clusters are relative to their cohesion. It is a useful diagnostic, not proof that the segmentation is commercially optimal.

6. Segment interpretation

Review the generated segment_profile.csv file for average age, income, spending score, purchase frequency, order value and recency. Segment IDs are arbitrary machine-generated labels; interpret them from their measured characteristics rather than treating a larger ID as a better segment.

7. Business recommendations

Retention: target long-inactive customers with win-back campaigns. Loyalty: test exclusive benefits for frequent, high-value customers. Personalisation: tailor product recommendations to segment behaviour. Discount strategy: use discounts selectively and measure margin impact. Measurement: monitor repeat purchase rate, revenue per customer and campaign conversion by segment.

8. Limitations

Synthetic data does not represent actual customer behaviour. K-Means is sensitive to feature selection, scaling and cluster assumptions. Silhouette score alone cannot determine business usefulness. Transaction-level product categories, margins and lifetime revenue are not included.

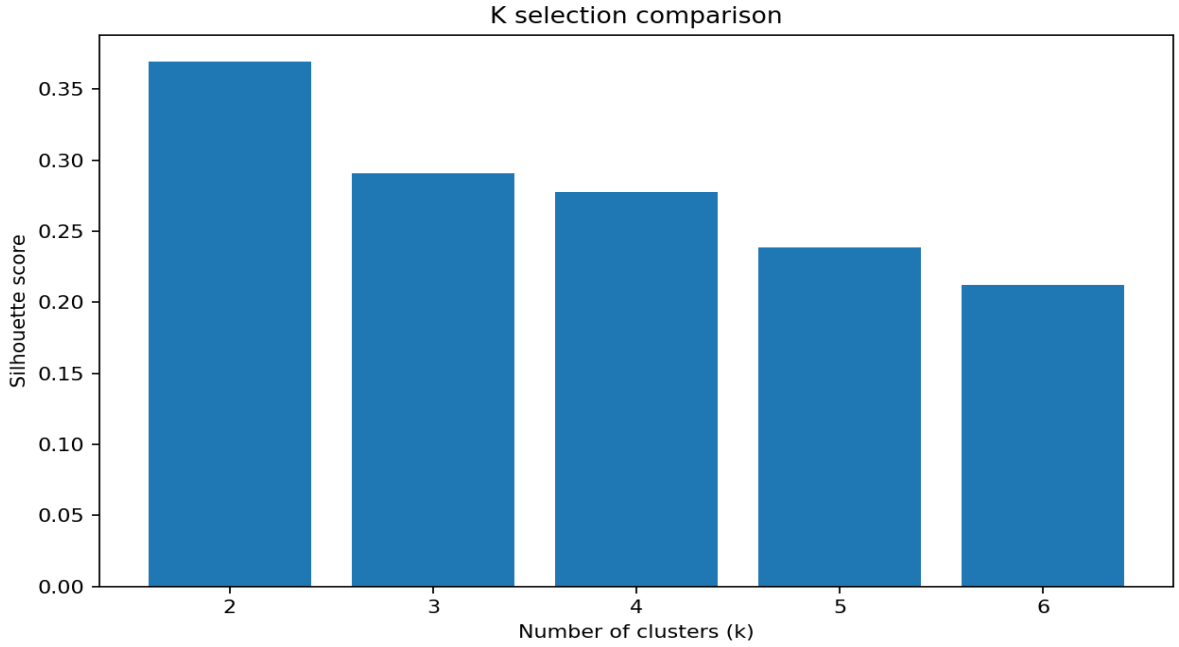
9. Conclusion

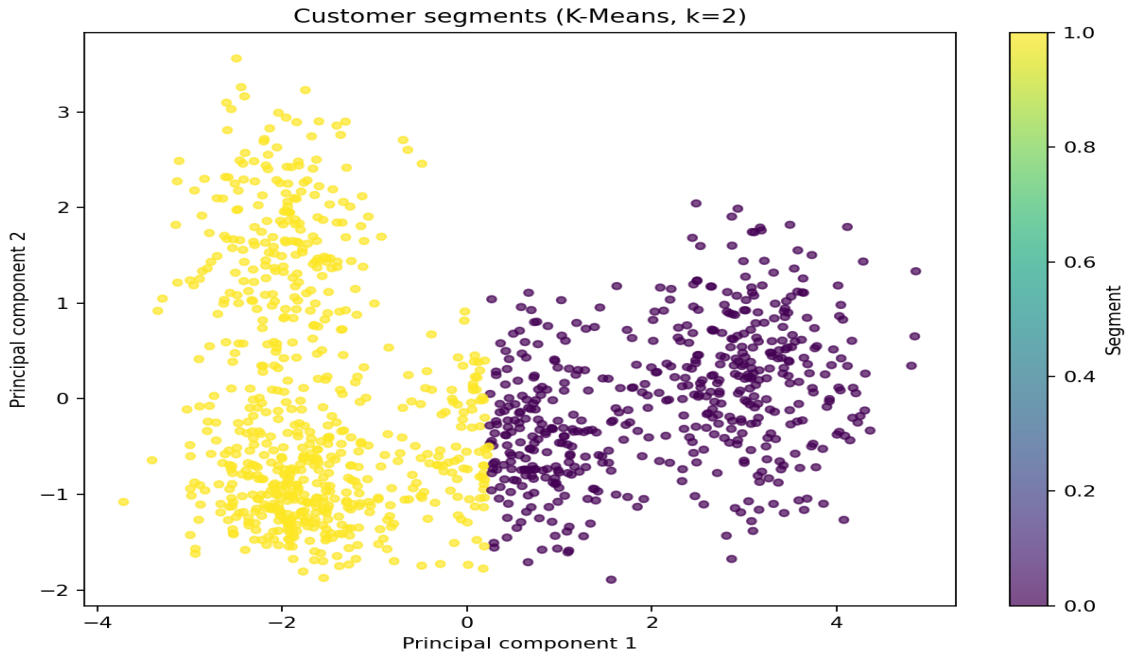
The project demonstrates an end-to-end customer segmentation workflow. For real business use, rerun it on approved anonymised data, validate the segments with business stakeholders and monitor whether

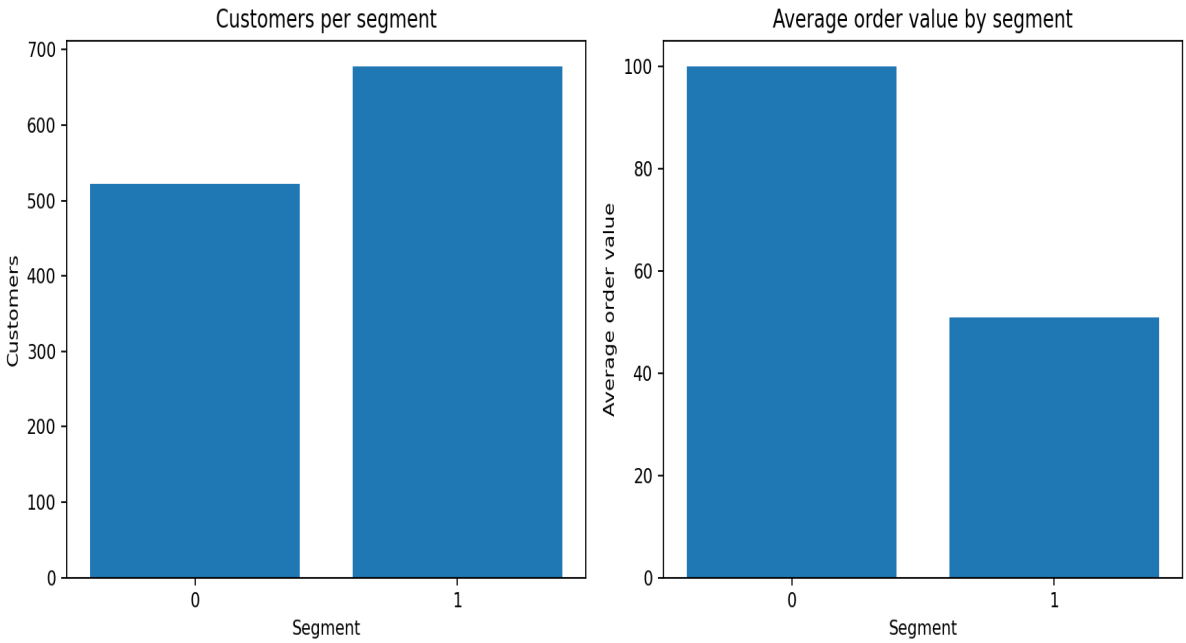
segment-specific actions improve measurable outcomes.

10. Results and visualisations

Segment	age	annual_income_k	spending_score	purchases_last_12m	avg_order_value	days_since_last_purchase	website_visits_monthly	discount_usage_rate
0	34.35	68.29	66.35	6.48	100.11	22.8	6.98	0.29
1	32.88	38.07	28.28	2.06	50.98	72.2	2.5	0.67






Customers per segment
Customers
Segment
0
1
0
100
200
300
400
500
600
700
Average order value by segment
Average order value
Segment
0
1
0
20
40
60
80
100